

CMSC 104 Section 2

Spring 2026

Answer Key for

Sample Quiz for Quiz 1

Overview:

This sample quiz is intended to prepare students for the actual quiz that will be taken in class on Wednesday, February 18. There are the same number, type, and point values as there will be on the actual quiz. Students will be given 75 minutes to complete the quiz on February 18, so that can be a guide in working through this sample quiz.

Instructions:

This quiz is to be completed on paper, using a pen or pencil. Please make sure you write your name on the paper. When you are finished you may turn in your paper and leave the room.

This quiz is “open-book, open-notes.” You MAY use any of the following: your laptop or phone, a book; your notes; the instructor’s notes and code from the GitHub site; an IDE such as IDLE or PyCharm (if you wish to test your code to make sure your answer is correct). You MAY NOT use: any other person, whether in this classroom or not (e.g., you may not text your friend the computer science major), or any AI-support source such as ChatGPT, Gemini, CoPilot, Claude Code, etc. If you have a question about this, please raise your hand and ask.

Section 1: True/False and Multiple Choice (40 points)

There are 10 questions in this section. Each is worth 4 points. No partial credit is given for any question in this section. If you change your answer during the quiz, please make sure you clearly indicate which is your final answer.

1. Which of the following is NOT a legal Python variable name?
 - a. `_name`
 - b. **`-name`**
 - c. `Name`
 - d. `N2me`

2. True or False: in Python the value of `1//3` is 0
- a. **True**
 - b. False
3. Which of the following would qualify as “system software?”
- a. **Microsoft Windows 11**
 - b. Microsoft Excel
 - c. A video game
 - d. Google Sheets
4. What’s the difference between `print(“UMBC”, “Retrievers”)` and `print(“UMBC”+”Retrievers”)`?
- a. There is no difference; they produce the same results
 - b. The first statement produces a blank space between UMBC and Retrievers while the second statement places a tab between them
 - c. The second statement produces a blank space between UMBC and Retrievers while the first statement runs them together with no blank space in between.
 - d. **None of the above is correct**
5. True or False: when your program includes the statement
- ```
s = input(“Please enter a string now”)
```
- The program waits until the user hits the “enter” (or “return”) key, Everything before that keystroke is part of the string value assigned to s.
- a. **True**
  - b. False
6. Which of the following would be considered computer hardware, as opposed to software:
- a. A keyboard
  - b. A monitor
  - c. A mouse
  - d. **All of the above are hardware**
7. True or False: the Python symbol table is a data structure that contains: the name of each symbol (variable) used in the program, plus its type, and the current value of that variable.
- a. True
  - b. **False**
8. If you wanted to prompt the user to enter an integer into your program, and you wanted to do math on what the user typed, which of the following statements would work?
- a. `S = input(“Please enter an integer”)`

- b. S = int(input("Please enter an integer"))**
  - c. S = str(input("Please enter an integer"))
  - d. All of the above would let you do math on the user's input
- 9. If the Boolean variable x = True, what is the value of not x?
  - a. True
  - b. False**
  - c. We don't know; it's undefined
  - d. All of the above are correct

10. True or False: the Python code
- ```
product = 1
for i in range(1,5,1):
    product = product * i
print(product)
```

Will result in the value 120 being printed on the screen.

- a. True
- b. False**

Section 2: Short answer (40 points)

There are 8 questions in this section. Each is worth 5 points. Partial credit IS given for questions in this section, so please make an attempt to answer each question.

Note: "short answer" means one or two sentences. It should not take you more than that to answer a question. Continuing to write paragraph after paragraph is probably not going to help make your answer more correct.

11. What does this code result in?
- ```
for i in range(1,5,1):
 product = product * i
print(product)
```

**An error, because product has no initial defined value**

12. A for i loop has the syntax
- ```
for i in range(start, stop, step):
    #some code to execute
```

What does the step value mean?

How much you increment i by each time the loop executes

13. Why is the variable name `2cool4school` illegal in python?

Variables cannot start with digits

14. Which type of error - syntax or semantic - will result in a program running, but not giving the output the programmer expected or wanted?

Semantic

15. Why would a programmer want to import the Python math library rather than writing all the code directly each time?

It's cheaper and easier to use code that somebody else has already written and validated. And there's less likely to be a bug in the code.

16. What is the value of

`(5*3//2)**(4//3)**(4%3)`

When evaluated by Python?

7 Evaluate the terms inside the parentheses first, and you have 71**1 or 7**1 which is just 7**

17. Suppose that $x = \text{True}$, $y = \text{False}$ and $z = \text{True}$. What is the value of $\text{not}(x \text{ or } y \text{ and } z)$?

False

18. Give one example of computer hardware, and one example of computer system software.

Hardware: a mouse

Software: Windows, MacOS, Android,...

Section 3: Programming (20 points)

There are two questions in this section; each is worth 10 points. Partial credit IS given for questions in this section.

You do not need to write comments with your code, unless you think that it will help the grader more clearly understand exactly what you're trying to accomplish.

19. Write a program that asks the user to input an integer, and then prints out the number that is the cube root of that number. You can assume that the user will input an integer - no error checking is needed. Please include some sort of label in the "print" statement that tells the user what you're printing out.

```
num = int(input("Please enter an integer and we will find its cube root"))  
print("The cube root of ", num, " is ", num**(1/ 3))
```

(You could also import the math library and use math.cbrt())

20. Write a program that asks the user to input an integer greater than 10, and then calculates the sum of all the numbers from 11 up to that number. (Include both 11 and the user's number in your sum!!) Use a "for i" loop to do this. Print out the answer, and make sure that the answer is labeled so that the user understands what she's looking at.

```
num = int(input("Please enter an integer greater than 10; we will calculate  
the sum of the numbers from 11 to your number, inclusive"))  
sum = 0  
for i in range(11, num+1, 1):  
    sum = sum + i  
print("The sum of the numbers from 11 through your number is ", sum)
```